Aditya Joshi

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Aditya Joshi Adi-j-11

Profile

Highly motivated Electronics and Communication Engineering student with a solid foundation in fundamental concepts and a deep passion for technology. Eager to apply knowledge and skills in real-world settings, help advance technological growth and create in impact in the field of science and technology. Looking forward for opportunities to apply and develop my skills in the field of Embedded Systems and Computer Architecture

Education

R.V College of Engineering, Bengaluru *2021 - 2025*
B.E in Electronics and Communication Engineering

- **GPA**: 8.8
- **Activities:** Quiz Corp and Debating Society, Dramatics, IEEE

Delhi Public School *2019 -2021*
Grade XI - Grade XII

- **Score**: 96 %
- **Activities:** Gavel Club, Atal Tinkering Lab

Experience

Firmware Intern *Bengaluru*
NXP Semiconductors *Jan 2025 – June 2025*

- Incorporated Endurance Testing into the existing test framework for Firmware validation of NXP's Hardware Security Module(HSM)
- Developed FreeRTOS and NXP RTOS based frameworks for validation and testing of Hardware Security Modules (HSMs)
- Developed an automated validation framework using Python and PE Micro tools for RTOS based test configuration.

Electronics Lead *Bengaluru*
Team Antariksh *April 2022 – Dec 2024*

- Lead the Electronics Control and Logic Subsystem for the design and development of the RVSAT-1 India's first microbiological payload built for the POEM Platform and launched on the PSLV C-60 Spadex mission (Dec 2024)
- Designed the Flight Computer for RVSAT-1 which includes communication interface and controls for the payload
- Designed a modular photospectrometer for the measurement of Microbial growth for the Payload
- Designed a Electrical Power System and converter system for RVSAT-1

Project Intern *Bengaluru*
Wipro IISC Research Network *Nov 2023 – April 2024*

- Performed a study on ECU units and Architecture of Autonomous Vehicle and contributed to the design and organization of ECU units
- Designed and implemented a Data Acquisition Unit for autonomous vehicles to collect sensor data in real time using Raspberry Pi and STM32
- Designed a Vehicle Health Management system using CAN Protocol, STM32 and Raspberry Pi
- Implementation of of CAN Protocol on SPC5 automotive grade Microcontroller

Projects

Classification and Scheduling Framework for Heterogeneous computing *2025*

- Designed a simple framework to extract computationally intensive segments of a program using LLVM compiler toolchain
- Integration of a ILP based scheduling algorithm to allocate tasks on a heterogeneous computing platform (CPU+GPU) and improved overall execution effeciency

FPGA Based Edge Computing Platform *2024*

- Designed a FPGA based platform for real time image processing from wireless camera node
- Tools Used: Raspberry Pi, Zynq 7000 FPGA, Verilog

Sanitary Pad Vending Machine *2023*

- Developed an IOT enabled high capacity Sanitary Pad Vending Machine **(Design Patent filed)** for Industrial purpose
- Tools Used: ESP32, MQTT, Altium

Technologies

Languages: C, C++, Python, ARM Assembly

Technologies: Raspberry Pi, Cortex M4, FPGA, STM32, LLVM

Achievements and Extracurricular Activities

Smart India Hackathon 2024 Grand Finalists

- Represented RVCE at the 5 day long Grand Finals of SIH 2024 Hardware track
- The team developed a prototype of a smart respiratory monitoring system for athletes, integrating real-time sensor data acquisition and analysis for performance tracking

Member of RV Quiz Corp and Debating Society

- Participated and represented RVCE in Quzzing and Debating competitions across India
- Member of Organizing committee for UTPT (Quizzing Fest) and RVDT (Debating Tournament)

Merit Award 2025

- Received Merit award from for the Launch RVSAT-1
- Merit award for Representing RVCE as finalists in Quiz Competition